

ceptional ways, provided we choose m large enough and ϵ small enough (giving an additional upper bound for ϵ). It is now a routine matter to compute the values of m and ϵ that will work.

7.6 Two regularity tools

Our acquaintance with the regularity lemma so far already includes the essentials: the notion of regularity, a statement of the regularity lemma, its proof, and a typical application of the lemma in our proof of Theorem 7.1.2 in Section 7.5. Another application, also based on no more than we have covered so far, will be given in the proof of Theorem 9.2.2. Both these applications illustrate techniques typically involved in many uses of the regularity lemma.

Our aim in this section is to extract some of these techniques and bundle them into formal tools that can be used out of the box: the ‘counting lemma’, the ‘removal lemma’, and some immediate consequences of these. Their proofs will not be short, but the ideas they involve will look familiar by now. Conversely, our proofs of Theorems 7.1.2 and 9.2.2 could be rewritten to use some of these formal tools (Exercise 45).

To facilitate the proof of the counting lemma, it will be convenient to use a slightly different notion of ϵ -regularity. To avoid confusion, we shall call this ‘ ϵ -uniformity’ here; but in the literature it is usually referred to as (another version of) ‘ ϵ -regularity’ as well.

Call a pair (A, B) of disjoint sets of vertices in a graph (ϵ, d) -uniform for $\epsilon > 0$ and $0 \leq d \leq 1$ if their non-empty subsets $X \subseteq A$ and $Y \subseteq B$ satisfy

$$|\|X, Y\| - d|X||Y|| \leq \epsilon|A||B|,$$

or equivalently,

$$|d(X, Y) - d| \leq \epsilon \frac{|A||B|}{|X||Y|}.$$

We may think of the real number d as a ‘target density’ with which the densities $d(X, Y) = \|X, Y\|/|X||Y|$ of the pairs (X, Y) are compared. If this is chosen as $d = d(A, B)$, the (ϵ, d) -uniform pair (A, B) is simply called ϵ -uniform.

This differs from our definition of ϵ -regularity in Section 7.4 in two ways. One is that all subsets $X \subseteq A$ and $Y \subseteq B$ are required to satisfy the inequality, including those of size less than $\epsilon|A|$ or $\epsilon|B|$. The other difference is best seen in the second of the two inequalities. Its left-hand side measures how much the density of (X, Y) differs from that of (A, B) . Our definition of ϵ -regularity requires that this should be at most ϵ . This is relaxed here by the factor $|A||B|/|X||Y| \geq 1$: the density of small pairs (X, Y) is allowed to deviate from d more than that of bigger ones.

(ϵ, d) -
uniform

ϵ -uniform

The difference between ϵ -regularity and ϵ -uniformity, however, is qualitatively irrelevant: ϵ -regular pairs are also ϵ -uniform, which we use freely in this section, and ϵ -uniform pairs are $\sqrt[3]{\epsilon}$ -regular (Exercise 42).

The purpose of the counting lemma is to count in how many ways a given graph H can be embedded as a subgraph, not necessarily induced, in a larger graph G that comes with an ϵ -regular partition. In our version we pre-assign to each vertex of H the partition class of G in which we want to embed it. We then count only those embeddings $H \rightarrow G$ that conform to this assignment, and assume that the pairs of partition classes due to accommodate the edges of H form ϵ -regular pairs in G . The lemma then says, roughly, that the number of such embeddings is close to what their expected number would be if G was a random graph that had edges only between distinct classes V_i, V_j and these edges were chosen independently with probabilities d_{ij} close to $d(V_i, V_j)$.¹⁰

It turns out that, for G large enough, there are at least $c|G|^{|H|}$ such embeddings $H \rightarrow G$, where c depends only on H , ϵ , and the densities $d(V_i, V_j)$ between the partition classes needed to accommodate the edges of H . If all these densities are positive we have $c > 0$, and can therefore find at least one copy of H in any large enough G that has such an ϵ -regular partition.

Before we state and prove the counting lemma, let us note what happens in the special case that $H = K^k$ and G is k -partite with classes $V_1, \dots, V_k$, for some $k \in \mathbb{N}$. Since the vertices of H are pairwise adjacent, any embedding in G sends them to distinct classes, and there are $\prod_{1 \leq i \leq k} |V_i|$ ways of doing that. Not all of them will induce a K^k in G , but we can estimate how many of them might: a pair (v_i, v_j) from (V_i, V_j) induces an edge in G with a ‘probability’ of $d(V_i, V_j)$. If these are close to given parameters d_{ij} , we can expect that about $\prod_{1 \leq i < j \leq k} d_{ij}$ of our $\prod_{1 \leq i \leq k} |V_i|$ maps $V(K^k) \rightarrow V(G)$ induce a K^k also in G .

Proposition 7.6.1. *Let $\epsilon > 0$, and let G be a multipartite graph with vertex classes $V_1, \dots, V_k$ whose pairs (V_i, V_j) are (ϵ, d_{ij}) -uniform in G . Then*

$$\left| |\mathcal{K}^k(G)| - \prod_{1 \leq i < j \leq k} d_{ij} \cdot \prod_{1 \leq i \leq k} |V_i| \right| \leq \epsilon \binom{k}{2} \prod_{1 \leq i \leq k} |V_i|,$$

where $\mathcal{K}^k(G)$ denotes the set of complete subgraphs of order k in G .

Proposition 7.6.1 will follow from Lemma 7.6.2 (Exercise 44).

For the more general counting lemma we need some notation. Let us denote by $\text{Hom}(H, G)$ the set, and by $\text{hom}(H, G)$ the number, of homomorphisms from H to G . Recall that these are the maps $V(H) \rightarrow V(G)$

$\text{Hom}(H, G)$
 $\text{hom}(H, G)$

¹⁰ Both uses of ‘close’ here mean, essentially, ‘up to an ϵ -portion’.

that preserve adjacency, but not necessarily non-adjacency. Vertices of H can map to the same vertex of G , but only if they are non-adjacent.¹¹

Injective homomorphisms $\varphi: H \rightarrow G$ embed H in G as a subgraph, though not necessarily as an induced subgraph. Note that there can be more such homomorphisms than subgraphs of G isomorphic to H , because there can be many ways of mapping H to a fixed copy of H in G . We may think of each φ as specifying a ‘labelled copy’ of H in G : one whose vertices are each ‘labelled’ by the vertices of H that φ maps there.

Consider a fixed homomorphism $\varphi: H \rightarrow R$ with $V(R) = \{1, \dots, k\}$, and let G be a k -partite graph with vertex classes $V_1, \dots, V_k$. Given any subgraph $H' \subseteq H$ (often, but not always, $H' = H$) we write $\text{Hom}_\varphi(H, G)$ $\text{Hom}_\varphi(H', G)$ for the set, and $\text{hom}_\varphi(H, G)$ $\text{hom}_\varphi(H', G)$ for the number, of homomorphisms $\psi: H' \rightarrow G$ that satisfy $\psi(u) \in V_{\varphi(u)}$ for all $u \in V(H')$; we call these the φ -partite homomorphisms from H' to G .¹²

Lemma 7.6.2. (Counting Lemma)

Let R be a graph on $\{1, \dots, k\} \subseteq \mathbb{N}$, and $\varphi: H \rightarrow R$ a homomorphism for some graph H . Let G be a multipartite graph with vertex classes $V_1, \dots, V_k$, whose pairs (V_i, V_j) with $ij \in E(R)$ are (ϵ, d_{ij}) -uniform in G for some $\epsilon > 0$ and various $d_{ij} \geq 0$. Then

$$\left| \text{hom}_\varphi(H, G) - \prod_{e \in E(H)} d_e^\varphi \cdot \prod_{u \in V(H)} |V_{\varphi(u)}| \right| \leq \epsilon \|H\| \prod_{u \in V(H)} |V_{\varphi(u)}|,$$

where $d_e^\varphi = d_{\varphi(u)\varphi(w)}$ for edges $e = uw$ of H .

A typical application of the counting lemma, which we shall prove below, says that there are many φ -partite homomorphisms $H \rightarrow G$, at least

$$\text{hom}_\varphi(H, G) \geq \left(\prod d_e^\varphi - \epsilon \|H\| \right) \cdot \prod |V_{\varphi(u)}|$$

many, of which comparatively few are non-injective. The conclusion, then, is that there are many injective homomorphisms $H \rightarrow G$. In particular, there is at least one, so G contains H as a subgraph.

For example, if $\chi(H) = k$, say, let us show that we can find a copy of H in any graph G that has a vertex partition $\{V_1, \dots, V_k\}$ whose pairs (V_i, V_j) are ϵ -uniform with positive density, at least $d > 0$ say, as soon as the classes V_i have some common size ℓ that is large enough in terms of d and ϵ , and ϵ is small enough that $d^{\|H\|} - \epsilon \|H\| > 0$.¹³

¹¹ ... unless G has loops, which in this book happens only in Chapters 4.6 and 6.

¹² Fixing $\varphi: H \rightarrow R$ and then looking for injective φ -partite homomorphisms $H \rightarrow G$ is similar to finding embeddings $H \rightarrow G$ based on a specified embedding $H \rightarrow R_s$, as we did in our version of the blow-up lemma, Lemma 7.5.2.

¹³ In an application, these V_i might be only some of the partition classes of a bigger graph to which we have applied the regularity lemma. And only some of their pairs may be ϵ -uniform with density at least d , just enough pairs for $\varphi: H \rightarrow R$ to exist.

As $\chi(H) = k$, there exists a homomorphism $\varphi: H \rightarrow R := K^k$. We apply the counting lemma with $\epsilon < d^{\|H\|}/\|H\|$ and $d_{ij} := d(V_i, V_j) \geq d$. It tells us that

$$\text{hom}_\varphi(H, G) \geq (d^{\|H\|} - \epsilon \|H\|) \ell^{|H|}.$$

Let us think of this lower bound as $c \ell^{|H|}$ in which c is a ‘constant’: it depends only on H , d and our choice of ϵ (which ensures that $c > 0$), but not on ℓ . Our aim now is to show that if ℓ is large enough there are more injective homomorphism than non-injective ones.

How many non-injective φ -partite homomorphisms $\psi: H \rightarrow G$ are there? Any such ψ maps two vertices u, w of H to the same vertex of G , and for every choice of u and w there are at most $\ell^{|H|-1}$ such ψ . Hence for $c' = \binom{|H|}{2}$ there are at most $c' \ell^{|H|-1}$ non-injective φ -partite homomorphisms $H \rightarrow G$ altogether. As $c \ell^{|H|} - c' \ell^{|H|-1} \rightarrow \infty$ as $\ell \rightarrow \infty$, we find that for ℓ large enough there exists an injective φ -partite homomorphism $H \rightarrow G$. In particular, G has a subgraph isomorphic to H .

Proof of Lemma 7.6.2. We apply induction on $\|H\|$. The induction starts trivially with $\|H\| = 0$, where we follow the convention that the empty product $\prod_{e \in \emptyset} d_e^\varphi$ equals 1. For the induction step, pick an edge $ab \in H$ and let $H' := H - ab$.

The homomorphisms in $\text{Hom}_\varphi(H, G)$, which we wish to count, are precisely the homomorphisms in $\text{Hom}(H', G)$ that map a, b to adjacent vertices of G . Let us write $\mathbb{1}_G$ for the function that sends a pair of vertices of G to 1 if they are adjacent, and to 0 if not; in particular, $\mathbb{1}_G(v, v) = 0$ for all $v \in V(G)$. Then

$$\text{hom}_\varphi(H, G) = \sum_{\psi \in \text{Hom}_\varphi(H', G)} \mathbb{1}_G(\psi(a), \psi(b)).$$

In order to apply the induction hypothesis, we rewrite this sum as

$$\begin{aligned} \text{hom}_\varphi(H, G) &= \sum_{\psi \in \text{Hom}_\varphi(H', G)} \left(\mathbb{1}_G(\psi(a), \psi(b)) - d_{ab}^\varphi + d_{ab}^\varphi \right) \\ &= \sum_{\psi \in \text{Hom}_\varphi(H', G)} \left(\mathbb{1}_G(\psi(a), \psi(b)) - d_{ab}^\varphi \right) + d_{ab}^\varphi \cdot \text{hom}_\varphi(H', G), \end{aligned}$$

where d_{ab}^φ is defined as in the statement of the counting lemma.

Qualitatively, the induction hypothesis tells that the last term above, $\text{hom}_\varphi(H', G)$, is close to $\prod_{e \in E(H')} d_e^\varphi \cdot \prod_{u \in V(H')} |V_\varphi(u)|$. As $V(H') = V(H)$, this differs from the corresponding target value for $\text{hom}_\varphi(H, G)$ only by a factor of d_{ab}^φ : exactly the factor that precedes $\text{hom}_\varphi(H', G)$ in the equation above. It thus remains to show that the big sum in the last equation is small.

Quantitatively, the last summand $d_{ab}^\varphi \cdot \text{hom}_\varphi(H', G)$ in our rewriting of $\text{hom}_\varphi(H, G)$ differs from the target value for $\text{hom}_\varphi(H, G)$ as stated in the counting lemma, the number $\prod_{e \in E(H)} d_e^\varphi \cdot \prod_{u \in V(H)} |V_\varphi(u)|$, by

$$\begin{aligned}
 & \left| d_{ab}^\varphi \cdot \text{hom}_\varphi(H', G) - \prod_{e \in E(H)} d_e^\varphi \cdot \prod_{u \in V(H)} |V_\varphi(u)| \right| \\
 &= \left| d_{ab}^\varphi \cdot \text{hom}_\varphi(H', G) - d_{ab}^\varphi \cdot \prod_{e' \in E(H')} d_{e'}^\varphi \cdot \prod_{u \in V(H)} |V_\varphi(u)| \right| \\
 &\leq d_{ab}^\varphi \cdot \epsilon \cdot \|H'\| \cdot \prod_{u \in V(H)} |V_\varphi(u)| \\
 &\leq \epsilon \cdot (\|H\| - 1) \prod_{u \in V(H)} |V_\varphi(u)|.
 \end{aligned}$$

This difference is smaller than the difference of $\epsilon \|H\| \prod_{u \in V(H)} |V_\varphi(u)|$ that the counting lemma allows between $\text{hom}_\varphi(H, G)$ and its target value of $\prod_{e \in E(H)} d_e^\varphi \cdot \prod_{u \in V(H)} |V_\varphi(u)|$: smaller by the quantity of $\epsilon \cdot \prod |V_u|$ that corresponds to the -1 in the last line above.

To complete the induction step, it thus remains to show that this margin of $\epsilon \cdot \prod |V_u|$ can accommodate the first summand in our earlier estimate for $\text{hom}_\varphi(H, G)$, i.e. that

$$\left| \sum_{\psi \in \text{Hom}_\varphi(H', G)} \left(\mathbb{1}_G(\psi(a), \psi(b)) - d_{ab}^\varphi \right) \right| \leq \epsilon \cdot \prod_{u \in V(H)} |V_\varphi(u)|. \quad (*)$$

H''

For a proof of $(*)$, let $H'' := H - a - b$ be the subgraph of H obtained by deleting the vertices a and b . Then

$$\begin{aligned}
 & \left| \sum_{\psi \in \text{Hom}_\varphi(H', G)} \left(\mathbb{1}_G(\psi(a), \psi(b)) - d_{ab}^\varphi \right) \right| \\
 &= \left| \sum_{\psi' \in \text{Hom}_\varphi(H'', G)} \sum_{\substack{\psi \in \text{Hom}_\varphi(H', G) \\ \psi|_{H''} = \psi'}} \left(\mathbb{1}_G(\psi(a), \psi(b)) - d_{ab}^\varphi \right) \right| \\
 &\leq \sum_{\psi' \in \text{Hom}_\varphi(H'', G)} \left| \sum_{\substack{\psi \in \text{Hom}_\varphi(H', G) \\ \psi|_{H''} = \psi'}} \left(\mathbb{1}_G(\psi(a), \psi(b)) - d_{ab}^\varphi \right) \right|.
 \end{aligned}$$

The first summation here, over all $\psi \in \text{Hom}_\varphi(H', G)$, is thus performed in steps: we first sum over all the possible ways ψ' in which ψ can act on the vertices other than a and b , and then sum for each of these ψ' over all the ways in which we can extend it to a and b to obtain a homomorphism ψ from all of H' to G . In particular, the ψ in the last sum have to map a to vertices of G that are adjacent to all the previously chosen images $\psi'(u)$

of neighbours u that a has in $V(H'')$, and likewise for b :

$$\psi(a) \in V_{\varphi(a)} \cap \bigcap_{u \in N_{H'}(a)} N_G(\psi'(u)) =: W_a$$

and

$$\psi(b) \in V_{\varphi(b)} \cap \bigcap_{u \in N_{H'}(b)} N_G(\psi'(u)) =: W_b.$$

W_a
 W_b

Conversely, every pair in $W_a \times W_b$, chosen as $\psi(a)$ and $\psi(b)$, extends a given $\psi' \in \text{Hom}_{\varphi}(H'', G)$ to a homomorphism $\psi \in \text{Hom}_{\varphi}(H', G)$. Thus,

$$\begin{aligned} & \left| \sum_{\psi \in \text{Hom}_{\varphi}(H', G)} \left(\mathbb{1}_G(\psi(a), \psi(b)) - d_{ab}^{\varphi} \right) \right| \\ & \leq \sum_{\psi' \in \text{Hom}_{\varphi}(H'', G)} \left| \sum_{\substack{\psi \in \text{Hom}_{\varphi}(H', G) \\ \psi|_{H''} = \psi'}} \left(\mathbb{1}_G(\psi(a), \psi(b)) - d_{ab}^{\varphi} \right) \right| \\ & = \sum_{\psi' \in \text{Hom}_{\varphi}(H'', G)} \left| \sum_{\substack{w_a \in W_a \\ w_b \in W_b}} \left(\mathbb{1}_G(w_a, w_b) - d_{ab}^{\varphi} \right) \right| \\ & = \sum_{\psi' \in \text{Hom}_{\varphi}(H'', G)} \left| \|W_a, W_b\|_G - d_{ab}^{\varphi} |W_a| |W_b| \right| \\ & \leq \text{hom}_{\varphi}(H'', G) \cdot \epsilon |V_{\varphi(a)}| |V_{\varphi(b)}| \end{aligned}$$

by the $(\epsilon, d_{ab}^{\varphi})$ -uniformity of $(V_{\varphi(a)}, V_{\varphi(b)})$ for their subsets W_a and W_b . Since

$$\text{hom}_{\varphi}(H'', G) \leq \prod_{u \in V(H'')} |V_{\varphi(u)}|,$$

assertion (*) follows. $\square$

We now turn to our second regularity tool, the removal lemma. Qualitatively,¹⁴ the removal lemma says that if a large graph G is so sparse that most of the embeddings of the vertices of some fixed graph H in G fail to induce a copy (or supergraph) of H , we can delete *all* the copies of H in G by removing few of its edges:

Lemma 7.6.3. (Removal Lemma)

For every graph H and every $\varrho > 0$ there exist $\eta > 0$ and n_0 such that the following holds for all graphs G of order $n \geq n_0$: if there are at most $\eta n^{|H|}$ injective homomorphisms $H \rightarrow G$, then G contains a set E of at most ϱn^2 edges such that $H \not\subseteq G - E$.

¹⁴ Quantitatively: if only $o(|G|^{|H|})$ of the maps $V(H) \rightarrow V(G)$ induce a supergraph of H in G , we can remove all the copies of H in G by deleting $o(|G|^2)$ edges.

Informally, the removal lemma says that if at most an η -fraction of all the maps $V(H) \rightarrow V(G)$ are injective $H \rightarrow G$ homomorphisms, then all these can be destroyed by deleting only a ϱ -fraction of the possible edges on $V(G)$. We can wish for ϱ to be as small as we like, but have to pay for it by having a smaller η and a bigger n_0 returned by the lemma, so the assertion will apply to fewer G and work only if there are fewer injective $H \rightarrow G$ homomorphisms.

The proof of the removal lemma will be along the following lines. The first thing we shall note is that we can allow non-injective $H \rightarrow G$ homomorphisms too in our count: there are too few of those to matter. All we shall need is that there are no more non-injective homomorphisms than injective ones, which will be easy to prove; then the number of all of them together is still bounded by $2\eta n^{|H|}$.

The desired few edges of G that hit all the copies of H in it will be all the edges outside ϵ -regular pairs of partition classes of density at least ϱ , for any ϵ -regular partition of G with ϵ chosen small enough. An easy count will show that there are at most ϱn^2 such edges, which we delete. The interesting bit then is why that destroys *all* the copies of H in G .

This is where ϵ -regularity and the counting lemma come in. If there is even a single copy H_0 of H left in G , its edges must lie in ϵ -regular pairs of partition classes of density at least ϱ . By the counting lemma, however, this means that there is not just one copy of H in G but many: more than our assumptions allow. Indeed, the existence of H_0 as above means that there exists a homomorphism $\varphi: H \rightarrow R$, where R is the regularity graph for precisely those ϵ -regular pairs that accommodate H_0 . The counting lemma, then, concludes that there should be more than $2\eta n^{|H|}$ homomorphisms $H \rightarrow G$ if η is small enough in terms of ϱ . But, as we have seen, this is ruled out by our assumption about η .

Proof of Lemma 7.6.3. If H has no edges, we can make the premise of the lemma void by suitable choices of η and n_0 , regardless of the value of ϱ . If H has an edge but $\varrho \geq 1/2$, the conclusion of the lemma holds with $E = E(G)$. We shall therefore assume that $\|H\| \geq 1$ and $\varrho < 1/2$.

On input

$$\epsilon, m \quad \varrho \quad \epsilon := \min \left\{ \frac{1}{8}\varrho, \frac{1}{2}\varrho^{\|H\|/\|H\|} \right\} \quad \text{and} \quad m := \lceil 2/\varrho \rceil \quad (1)$$

M the Regularity Lemma 7.4.1 returns an integer M . We use this to set

$$\eta, n_0 \quad \eta := \frac{1}{5}\varrho^{\|H\|/(2M)^{|H|}} \quad \text{and} \quad n_0 := \max \left\{ M, \lceil ({}^{|H|}_2)/\eta \rceil \right\}. \quad (2)$$

Now let G be any graph of order $n \geq n_0$ as considered in the lemma: such that there are at most $\eta n^{|H|}$ injective homomorphisms $H \rightarrow G$.

It is easy to see – compare our discussion just before the proof of the counting lemma – that there are also at most

$$n^{|H|-1}({}^{|H|}_2) \leq n^{|H|}({}^{|H|}_2)/n_0 \leq \eta n^{|H|}$$

non-injective homomorphisms $H \rightarrow G$. The number of all homomorphisms $H \rightarrow G$, injective or not, is therefore at most

$$\text{hom}(H, G) \leq 2\eta n^{|H|}. \quad (3)$$

The regularity lemma yields an ϵ -regular partition $\{V_0, V_1, \dots, V_k\}$ of G with $|V_1| = \dots = |V_k| := \ell$ and

$$2/\varrho \underset{(1)}{\leq} m \leq k \leq M. \quad (4)$$

Let us use this partition to choose our set E to satisfy the lemma. Let E be the set of all edges xy of G that satisfy one of the following statements:

- (a) x or y lies in V_0 ;
- (b) x and y lie in the same V_i with $i \geq 1$;
- (c) $x \in V_i$ and $y \in V_j$ with (V_i, V_j) not ϵ -regular and $i, j \geq 1$;
- (d) $x \in V_i$ and $y \in V_j$ with $d(V_i, V_j) < \varrho$ and $i, j \geq 1$.

To prove our lemma, we have to show that

$$|E| \leq \varrho n^2 \quad \text{and} \quad H \not\subseteq G - E. \quad (5)$$

For a proof of the first statement in (5) note that

- at most $|V_0| \cdot n \leq \epsilon n^2$ edges of G satisfy (a);
- at most $k \cdot \binom{n/k}{2} < n^2/2k$ edges of G satisfy (b);
- at most $\epsilon k^2 \cdot (n/k)^2 = \epsilon n^2$ edges of G satisfy (c);
- at most $\binom{k}{2} \cdot \varrho \cdot (n/k)^2 < \varrho n^2/2$ edges of G satisfy (d).

Hence

$$|E| \leq \left(\epsilon + \frac{1}{2k} + \epsilon + \frac{1}{2}\varrho\right)n^2 \underset{(4)}{\leq} \left(\epsilon + \frac{1}{4}\varrho + \epsilon + \frac{1}{2}\varrho\right)n^2 \underset{(1)}{\leq} \varrho n^2$$

as desired.

For a proof of the second statement in (5) suppose that $H \subseteq G - E$. By our choice of E , every edge of this copy of H lies in some ϵ -regular pair of partition classes of density at least ϱ . This defines a homomorphism $\varphi: H \rightarrow R$, where R is the regularity graph of G for our partition with parameters ϵ , ℓ and ϱ .¹⁵

¹⁵ The regularity graph for an ϵ -regular partition is defined before Lemma 7.5.2.

Since ϵ -regularity implies ϵ -uniformity (Exercise 42), φ and $G - E$ satisfy the premise for the counting lemma with $d_{ij} := d(V_i, V_j) \geq \varrho$ for the edges ij of R . The counting lemma yields

$$\begin{aligned}
 \text{hom}(H, G) &\geq \text{hom}_\varphi(H, G - E) \\
 &\geq \left(\prod_{e \in E(H)} d_e^\varphi - \epsilon \|H\| \right) \cdot \prod_{u \in V(H)} |V_{\varphi(u)}| \\
 &\geq (\varrho^{\|H\|} - \epsilon \|H\|) (n/2M)^{|H|} \\
 &\stackrel{(1)}{\geq} \frac{1}{2} \varrho^{\|H\|} (n/2M)^{|H|} \\
 &\stackrel{(2)}{>} 2\eta n^{|H|}.
 \end{aligned}$$

This contradicts (3), and thereby completes the proof of (5). $\square$

The number of copies of a fixed graph H in some dense graphs G tells us something about the structure of those G when they get large. Assume, for simplicity, that we have an infinite class of graphs G each with αn^2 edges, where $n = |G|$ and $\alpha > 0$ is some fixed parameter telling us how dense all those G are. Let $k := |H| \geq 3$ and $m := \|H\| \geq 1$. Let $\gamma > 0$ be any constant, and assume that the number $\theta = \theta(G)$ of copies of H in our graphs G satisfies

$$\gamma n^2 \leq \theta m \leq \eta n^k \quad (*)$$

for any given $\eta > 0$ when n is large enough.¹⁶ On average, every edge of G lies in $\theta m / \alpha n^2$ copies of H . The removal lemma tells us that we can hit all those copies of H with only ϱn^2 edges, for any $\varrho > 0$ we may choose, when n is large enough. These special edges each lie in at least $\theta / \varrho n^2$ copies of H on average: more than our overall average by a factor of at least $c := \alpha / \varrho m$. And since we may choose ϱ , while α and m are constant, we can make c as large as we like, just by making ϱ small and n large enough that the removal lemma applies.

So the removal lemma tells us that the copies of H in G cannot be spread evenly over G , in the sense that the maximum number of copies in which a given edge of G lies differs from the average such number by no more than a constant factor as n grows. The copies of H in G must instead be lumped together at a few edges that lie in many more such copies than an edge of G does on average: in at least c times as many, where $c \rightarrow \infty$ as $n \rightarrow \infty$. In absolute terms, these special edges lie in at least

$$c \cdot \theta m / \alpha n^2 \geq c \gamma / \alpha$$

¹⁶ In complexity-theoretic notation: $\theta(G) = \Omega(n^2)$ but $\theta(G) = o(n^k)$. The η here could be any returned by the removal lemma on input some ϱ we may later choose.

n, α
 k, m
 γ, θ

c

copies of H on average, for n large enough. Since γ/α is a positive constant, this number will get arbitrarily large if we choose ϱ small enough. This bites most if the average number $\theta m/\alpha n^2$ of copies of H in which an edge of G lies is itself bounded as n grows. We shall return to this case in a moment.

By contraposition, our considerations above also tell us something about the density of the graphs G in a given class, and about the number of copies of H they contain, if those copies are *not* lumped together at a few edges but are evenly spread over G . If these G are dense, we must also have $\theta \geq \gamma n^2$ for some $\gamma > 0$ (unless $\theta = 0$), since otherwise the copies of H would technically be ‘lumped together’ (as defined earlier) at every edge of G that lies in at least one of them. So our graphs satisfy the left inequality in (*), which means that the right one must fail: they must contain at least ηn^k copies of H for some fixed $\eta > 0$. In particular, for some mysterious reason exposed by the removal lemma, we *cannot have quadratically*¹⁷ many copies of H spread evenly over dense graphs G .

Let us say that the copies of H are *spread thinly* over the graphs G in some infinite class $\mathcal{G}$ if, for every $G \in \mathcal{G}$, every edge of G lies in at least one copy of H in G but only in a bounded number: in at most d copies, say, with the same $d \in \mathbb{N}$ for all $G \in \mathcal{G}$. Then θ and $\text{hom}(H, G)$ grow at most quadratically in n , more slowly than ηn^k for any η returned by the removal lemma. Hence those G will satisfy the premise of the removal lemma as soon as they are big enough. Our considerations above imply that those G cannot be dense. Let us show this more formally:

*spread
thinly*

Theorem 7.6.4. *Let H be any graph with at least 3 vertices. Let $\mathcal{G}$ be a class of graphs over which their subgraphs isomorphic to H are spread thinly. Then for every $\delta > 0$ there exists an integer n_0 such that every $G \in \mathcal{G}$ of order $n \geq n_0$ has at most δn^2 edges and contains at most δn^2 copies of H .*

Proof. Let $k := |H| \geq 3$ and $m := \|H\|$; we may assume that $\|H\| \geq 1$. Let d be the constant from the definition of ‘thinly spread’: an integer such that every edge of any $G \in \mathcal{G}$ lies in at most d copies of H in G . For each G , write $h = h(G)$ for the number of injective homomorphisms $H \rightarrow G$.

k, m
 d
 h

We apply the removal lemma with $\varrho = \delta/(m k! d)$. It returns some $\eta > 0$ and an integer ‘ n_0 ’, which we now call n_1 . We use these to define

ϱ, η

$$n_0 := \max \{n_1, \lceil k! \cdot d/\eta \rceil\}.$$

Let $G \in \mathcal{G}$ of order $n \geq n_0$ be given. Each copy of H in G is the

¹⁷ In complexity-theoretic notation: $\Theta(n^2)$ many.